MAXIMILIAN COMFERE

New York, NY | (507) 250-6563 | mkc2182@columbia.edu

EDUCATION

Columbia University, Fu School of Engineering and Applied Science — New York, NY
Bachelor of Science, Computer Engineering — Expected May 2026

PROFESSIONAL EXPERIENCE

Department of Physiology and Biomedical Engineering, Mayo Clinic — Rochester, MN
Engineering Co-op Intern — June 2025 – August 2025

- Developed a **multi-classifier ML pipeline** and **CNN** architectures in **PyTorch,** achieving **94% accuracy** in classifying photoplethysmography (PPG) waveforms on a 0–5 quality scale.
- Proposed a novel construction: Large Physiological Signal Models (LPSMs), leveraging transformer architectures and Llama 3.1 to capture long-range waveform dependencies and integrate temporally aligned HER textual context.
- Built a Python package for waveform annotation with a **FastAPI backend**, enabling thread-safe multi-user labeling and real-time training on up to **12 concurrent data streams**.

Columbia's Mortimer B. Zuckerman Mind, Brain and Behavior Institute — New York, NY
Undergraduate Software Lab Assistant — September 2023 – August 2024

- Implemented MoSeq, a **motion sequence machine learning model**, to observe and then map behavioral characteristics of Naked Mole Rats to specific castes within the colony.
- Designed large-scale **transition matrices** using Python to create visual representations of behavioral sequences.
- Modeled a tunnel game in Python to efficiently pair all Naked Mole Rats against each other to determine a dominance hierarchy for the colony.

Center for Digital Health, Mayo Clinic — Rochester, MN
Software Engineer Intern — June 2023 – August 2023

- Automated synthetic test data generation for a myelodysplastic syndrome (MDS) risk assessment pipeline using **Python, Pandas**, and OpenAI's API, reducing manual **QA time by ~40%**.
- Built a secure internal team portal with **React, TypeScript,** and **Azure App Services**, centralizing workflows, documentation, and model outputs.
- Designed and tested front-end mechanics for a clinical risk calculator, integrating **RESTful APIs, SQL Server streams**, and **Azure cloud** services to deliver real-time predictions.
- Collaborated with data scientists to deploy a machine learning model that analyzed structured EHR and laboratory data to support clinician decision-making at point of care.

EXTRACURRICULARS

Lion Health, Columbia University — New York, NY
Head of Software Engineering — April 2024 – present

- **Lead 8 software engineers** who work in cross-disciplinary teams made of clinicians and hardware engineers to build, deploy and then present a range of biomedical solutions.
- Built a sleep disorder app that uses a **Core ML Model** fine-tuned to wake up patients experiencing involuntary manic and potentially dangerous episodes during sleep.

Track and field, Columbia University — New York, NY

- ***Honors****: NCAA USTFCCCA Division 1 All-Academic Team 2023* — September 2022 - present

SKILLS & INTERESTS

Computer: Java, C/C++, JavaScript, Swift, Python, React, Node, Swift, SQL, PyTorch, MATLAB, Typescript, Verilog, CSS, HTML
Language*:* English (native), German (advanced)
Interests: Writing, Premier League Soccer, Cycling, Backpacking, Cosmology